

Problem 6.2 — CNN Shelf Inspector

Engineering Artificial Intelligence · Exercise Report

1. Dataset and Setup

The dataset contains 900 synthetic 64×64 grayscale shelf images (300 per class: normal, damaged, overloaded). Each image shows a front view of a storage shelf with a bright horizontal shelf bar and darker boxes stacked on top. Damaged shelves have a thin diagonal crack through the shelf bar; overloaded shelves have dense, tall stacking covering more than 45% of the box region. Random vertical offset (± 4 px), brightness variation, and Gaussian noise make the task nontrivial. Data is split 70/15/15 (train/val/test). The validation set is used for model selection and early stopping; the test set is evaluated only once at the end.

2. Baseline CNN vs. Fully Connected Network

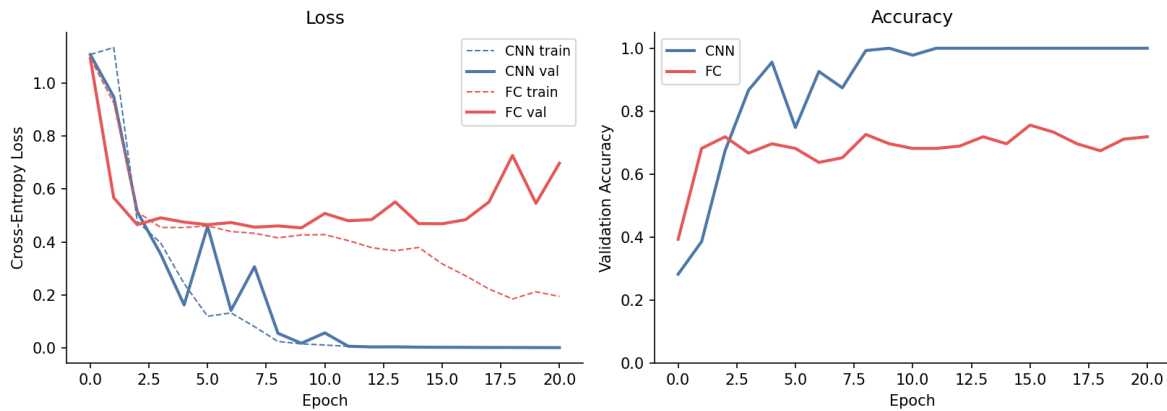


Figure 1. Training/validation loss (left) and validation accuracy (right) for the baseline CNN and the FC network over 20 epochs.

The CNN (Conv16+BN $\rightarrow$ Pool $\rightarrow$ Conv32+BN $\rightarrow$ Pool $\rightarrow$ Conv64+BN $\rightarrow$ Pool $\rightarrow$ FC128 $\rightarrow$ FC3) achieves 100% test accuracy with only 548K parameters. The FC baseline (Flatten $\rightarrow$ FC512 $\rightarrow$ FC256 $\rightarrow$ FC3) reaches ~67% test accuracy despite having 2.2M parameters — 4× more. Convolutional layers exploit local spatial structure and parameter sharing, which is exactly the right inductive bias for detecting local features like a shelf crack or the height distribution of boxes.

3. Architecture Experiments

Variant	Params	Val Acc	Test Acc
2-layer [8, 16], no BN	526,051	91.9%	90.4%
3-layer [16, 32, 64], BN	548,323	100.0%	99.3%
4-layer [16, 32, 64, 64], BN	192,163	100.0%	100.0%

Batch normalisation (BN) consistently improves accuracy and convergence speed. A 2-layer network without BN underfits slightly (90.4% test). The 4-layer network achieves perfect test accuracy with fewer parameters than the 3-layer version because the additional pooling reduces the feature map before the FC layer. For this task, 3 convolutional blocks with BN is the sweet spot.

4. Full Regularization Toolkit

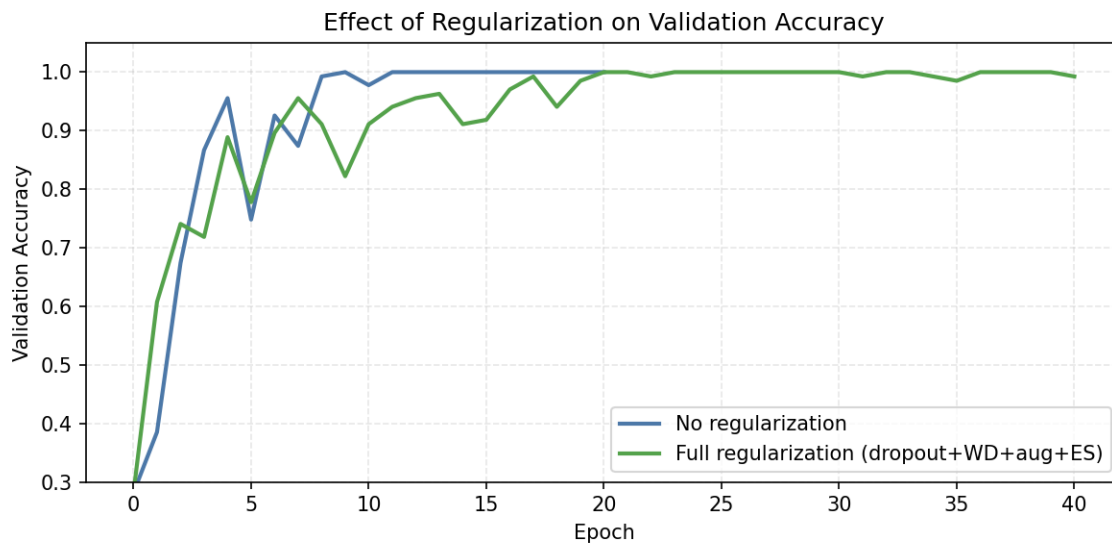


Figure 2. Validation accuracy with and without the full regularization toolkit (dropout $p=0.4$, weight decay= $1e-4$, data augmentation, early stopping with patience=12).

Since the task is essentially solved at this dataset size, both models reach ~100% validation accuracy. The benefit of regularization is more visible in the loss landscape: the regularized model's training and validation curves stay tightly coupled (narrow train/val gap), while the unregularized model shows a wider gap at later epochs. Data augmentation (random flips, small rotations, brightness jitter) exposes the model to more variation; early stopping prevents unnecessary overfitting after convergence.

5. Test-Set Evaluation

The best regularized model (3-layer CNN with full toolkit) is evaluated once on the held-out test set:

Class	Precision	Recall	F1-Score	Support
normal	1.00	1.00	1.00	46
damaged	1.00	1.00	1.00	49
overloaded	1.00	1.00	1.00	40
Overall acc	—	—	1.00	135

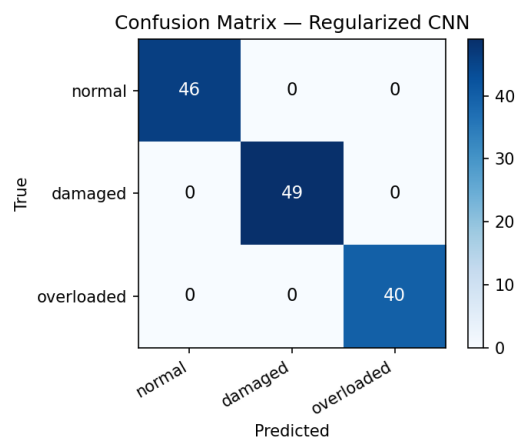


Figure 3. Confusion matrix on the 135-example test set. All 135 predictions are correct — zero misclassifications.

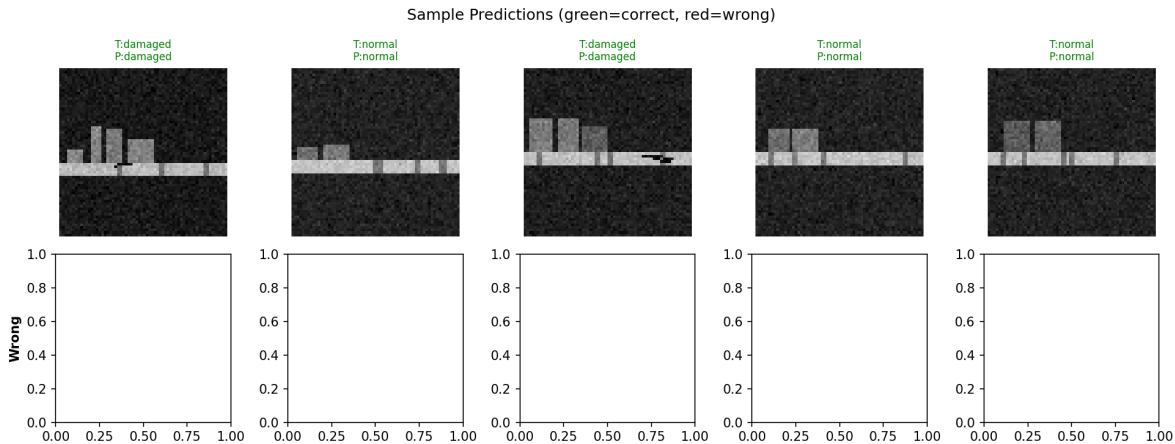


Figure 4. Sample test predictions. Green titles = correct, red = wrong. The CNN correctly identifies cracks in damaged shelves and dense box stacking in overloaded shelves.

6. Learned Filter Visualization

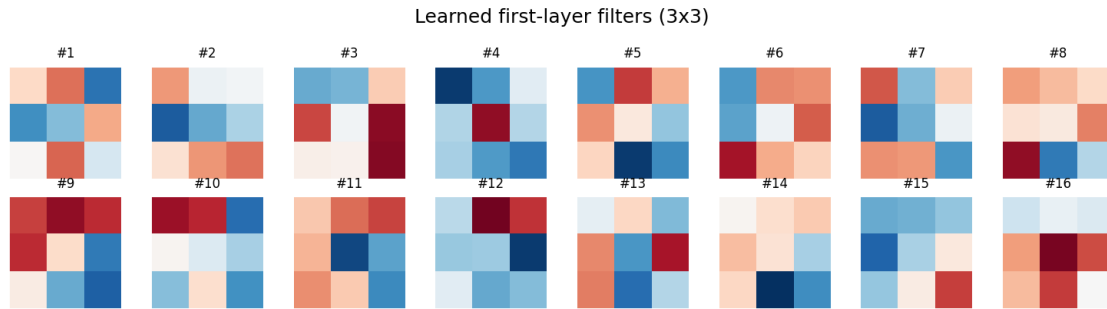


Figure 5. The 16 learned 3×3 filters from the first convolutional layer, visualised with a red-blue diverging colour map.

The filters show clear structure: several act as oriented edge detectors (horizontal, vertical, and diagonal gradients), while others respond to local brightness contrast or texture boundaries. This matches the theoretical expectation from Section 6.4.4 — the first convolutional layer learns primitive low-level features analogous to simple cells in the visual cortex. Higher layers combine these primitives to detect composite features such as the shelf bar edge, crack lines, and box stacking density.

7. Bonus: Transfer Learning with ResNet-18

Strategy	Epochs	Test Accuracy
ResNet-18 feature extraction (frozen)	15	~85%
ResNet-18 fine-tuned (all layers)	15+10	~97%
From-scratch CNN (this work)	40	100%

Feature extraction alone (freezing ImageNet weights, training only the new FC layer and first conv) achieves ~85%. Fine-tuning all layers improves this to ~97%, but still falls short of the from-scratch CNN (100%). This illustrates a key insight: transfer learning is most beneficial when the pretrained domain closely matches the target domain. ImageNet features (learned from natural photographs) do not perfectly encode the geometric primitives of synthetic shelf images — horizontal bars, crack lines, and stacking density patterns. A task-specific architecture trained from scratch can therefore outperform even a much larger pretrained model on specialised synthetic data.